Catalogue Studio

From a phone photo to a studio packshot

SARTIQ

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The task and our idea

Sartiq asked for a pipeline that turns amateur smartphone photos of a garment into a clean studio flat-lay packshot, the kind of image an online shop puts on a product page. Outputs are compared with a professional reference photo on colour (40%), perceptual similarity (25%), semantics (15%), silhouette (15%) and structure (5%).

Our idea is to *not* ask one generative model to do everything. The pipeline first understands the photo (which object is the product, where it is, which pixels belong to it), hands the generator a clean and honest reference, lets it do the one thing only it can do (re-pose, flatten, press and relight the garment), and then checks and corrects the result against the original photographs. The generator decides the shape; the photographs remain the source of truth for colour.

69.08
best dev leaderboard score, +19.2 in under six hours

≈80 s
per garment, end to end, fully automatic

6
hosted model calls per garment

1–5
photos per garment: flat, on a hanger or on a mannequin



Figure 1: The pipeline on a real dev garment (d023): five casual shots of a crumpled sweatshirt on a bed become one packshot. Every image is an actual intermediate output of the pipeline.

1 How it works

1 **Understand.** A vision-language model names the main product and writes a short description of it. A detector then finds the physical garment, so that a print on a T-shirt or a jacket hanging in the background is never mistaken for the product.

2 **Isolate.** A segmentation model selects every pixel of the product. The selection is trusted only when it agrees with the detector.

3 **Prepare references.** Instead of a hard cut-out, the original photo is cropped around the product with a small margin, so edges and fabric texture stay natural. Extra views (back, print close-up, label) are added as references; duplicates are removed.

4 **Generate.** An image-editing model produces one frontal, pressed flat-lay on a light studio background. The prompt asks it to keep colours, prints, lettering and every strap or button, and states that the photographs override the text description.

5 **Check and finish.** A vision model checks that exactly one garment is visible; a duplicate triggers one corrective edit and a mandatory recheck. The product is then matted, placed on a 1200×1800 canvas with its soft shadow, and a bounded colour correction pulls it back towards the colours measured in the source photos.

The steps are orchestrated as a graph (LangGraph) and every step has a fallback: an unusable detection falls back to the original photos, a missing matte keeps the generated image, and an inconclusive check marks the output for visual review. A garment never fails silently.

The whole run is automatic. A garment takes about 80 seconds and six hosted model calls (fewer when a fallback skips a step, at most eight when a duplicate must be repaired); generation alone is about a third of the time. The same pipeline runs from the command line, where all 42 garments of the challenge run in parallel with one command, and behind a web app that shows live progress and every intermediate image.

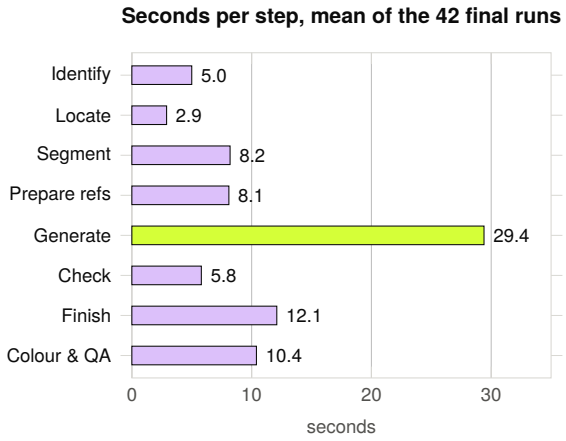




Figure 2: Phone photos (top) and final outputs (bottom) across very different capture conditions. g05 and g15 are on-site test garments shot on a hanger against a warm wooden wall.

2 Capture protocol

The pipeline is built to accept photos taken by anyone with a phone, but a few habits make the result noticeably more faithful.

- **One to five photos per garment.** The first photo is the front view and is the one the packshot reproduces. Extra photos of the back, of prints and of labels help the generator keep those details right.
- **The whole garment, with a margin.** Nothing should cover or cut the garment; other clothes in the frame are handled, but a clean frame is more reliable.
- **Any support works.** Flat on a surface, on a hanger or on a mannequin: the pipeline removes the support and flattens the garment. A smooth flat lay gives the most faithful silhouette.
- **Neutral, diffuse light.** Colour is taken from the photo, so avoid coloured or mixed light (a warm lamp, an orange wall, a window plus a bulb). A sheet of white paper in the frame is a cheap colour reference.
- **Get close enough for the details.** Prints and lettering survive only if they are sharp in at least one photo.

3 Results

In under six hours of iterations the dev score rose from 49.65 to 69.08 (Figure 3, left). On a local holdout the pipeline was level with the organisers' hosted baseline, within the run-to-run noise of about three points, but for a different reason: it reproduced shape and garment identity better and colour worse (right). The last iterations therefore went into colour fidelity.

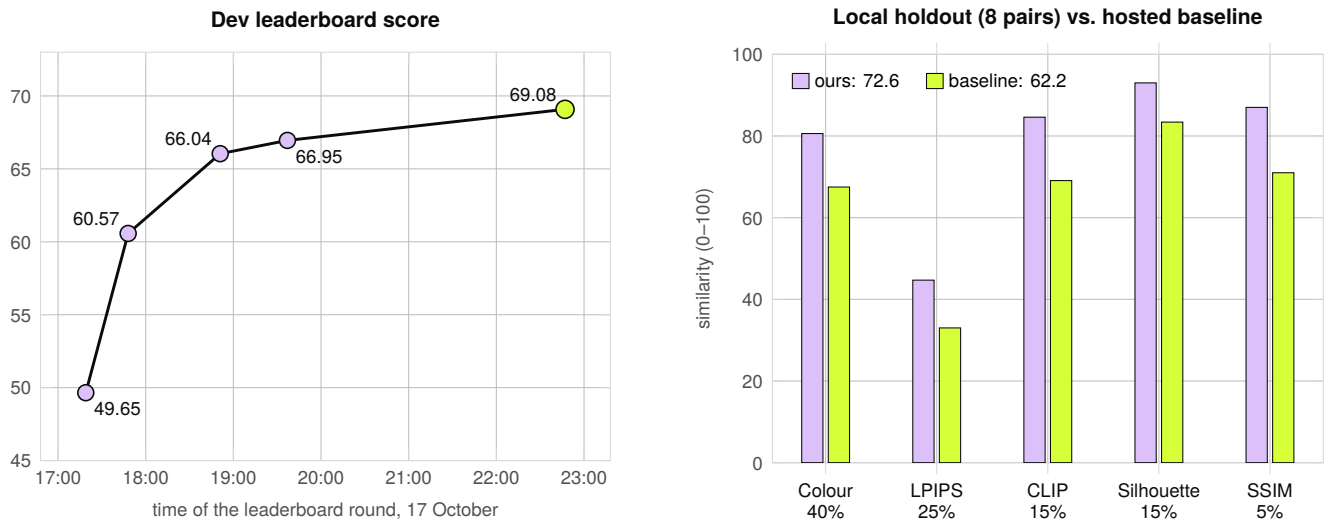
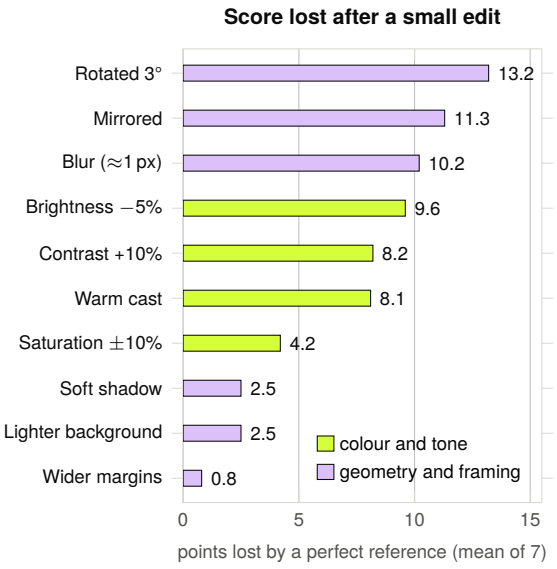


Figure 3: Left: our score on the live dev leaderboard during one afternoon of iterations. Right: our pipeline (17 October evening version) against the organisers' hosted baseline on a local holdout of 8 training pairs, per score component with its weight. It won on semantics and silhouette and lost on colour, which is what the later colour-fidelity work targeted. The holdout is a different set from the dev leaderboard, so its scores cannot be compared with the 69.08 on the left.

What we learned. Colour is where the points are. We took perfect references, perturbed them slightly and re-scored them (right): a light warm cast or a 5% change in brightness costs 7–10 points, as much as a visibly blurred image, while a lighter background, a soft shadow or wider margins cost at most 3. This is why the generator is asked to stay faithful rather than beautiful, and why the final step corrects colour against the source photos instead of simply whitening the background.

Generation is also noisy: the same garment moved by up to 20 points between two prompt variants, so we compared ideas on several garments instead of trusting single examples. The "pro" version of the image model scored worse than the standard one, because it restyled colours more freely.

Limitations. Generative editing can still alter small lettering or texture, and occasionally produces a duplicate that the check does not catch. A strong colour cast in the photo cannot always be undone, which is why the capture protocol insists on neutral light.



Running it. `validate_studio.py --views front.jpg [extra views]` produces one packshot plus its prompt, timings, trace and quality report; the code and the web app are in `dev/btc-sartiq/`.